

STAT 240

Week 11: Likelihood and Mixture Models

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Today

- Review: Regex and Strings
- Likelihood function
- Mixture Modeling

Logistics

- 2nd coding exam take home during final class April 10th
- Written exam April 14th (9.00am - 11am)
 - Will really be a 90 minute exam, but giving you longer
- Practice problems for written exam coming next week
- No class/lab next Friday on April 3rd (Easter)
 - Lab this week is a practice final coding exam

Course Evaluations

- Please fill them out!
- I'm curious for feedback on this course
 - Working on making this course as useful as possible

Quick Review of Regex and Strings

- Four key functions from `stringr`
 - Match a string and extract it: `str_extract` / `str_extract_all`
 - Detect a matching string `str_detect`
 - Replace one string with another `str_replace` / `str_replace_all`
 - Split a string on a substring `str_split`

Escaping and Anchors

```
mystrings <- c("abc", "a.c")  
str_extract(mystrings, ".")
```

```
[1] "a" "a"
```

```
str_extract(mystrings, "\\.")
```

```
[1] NA  "."
```

```
fruit[str_detect(fruit, "^a")]
```

```
[1] "apple"  "apricot" "avocado"
```

```
fruit[str_detect(fruit, "a$")]
```

```
[1] "banana"  "cherimoya" "feijoa"    "guava"     "papaya"    "satsuma"
```

```
fruit[str_detect(fruit, "^[t-z]")] [1:8]
```

```
[1] "tamarillo" "tangerine" "ugli fruit" "watermelon" NA  
[6] NA          NA          NA
```

```
mystrings <- c("$55.30", "$22.43")
str_extract(mystrings, "\\d") # Note that we need to "escape"
```

```
[1] "5" "2"
```

```
str_extract_all(mystrings, "\\d")
```

```
[[1]]
```

```
[1] "5" "5" "3" "0"
```

```
[[2]]
```

```
[1] "2" "2" "4" "3"
```

```
str_extract(mystrings, "\\$")
```

```
[1] "$" "$"
```

Working with Strings

- Working with strings is very much an exercise in trial and error
- It's important to know the key details but I don't know if anyone is actually "good" at using them
- But they are important, a basic competency is useful, makes you a much more efficient data scientist

Likelihood

What is Likelihood

- In statistics, a *model* is an assumption about how observed data arises
- For example, suppose we flip a coin N times. Assume that each flip results in H (heads) or T (tails). Let X_i be the result of the i -th flip
- Here's an example of a model for X_i :
 - The probability that the i -th flip results in heads is θ ($\theta \in [0, 1]$). The flips are independent events
 - This means the probability that the i -th flip results in tails is $1 - \theta$

Coin flip

- Under a model, the likelihood of the data (a function of the parameter) is the “probability” of the data given the parameter. For the coin flip model:

$$\mathcal{L}(\theta|X) = \Pr(X|\theta) \tag{1}$$

$$= \prod_{i=1}^N \theta^{I_H(X_i)} (1 - \theta)^{I_T(X_i)} \tag{2}$$

$$= \theta^{\#\{i: X_i = H\}} (1 - \theta)^{\#\{i: X_i = T\}} \tag{3}$$

- We will work backwards to see why it is this quantity

-
- Lets suppose we just have a single coin first
 - The probability that the flip results in heads is θ ($\theta \in [0, 1]$).
 - This means the probability that the i -th flip results in tails is $1 - \theta$
 - Can write the probability of the possible values of coin

$$\theta^k (1 - \theta)^{1-k},$$

where $k = 1$ if heads, $k = 0$ if tails

- This is the **probability mass function** (PMF)

Multiple Coins

- If we flip this coin several times, the result of the next flip won't depend on any previous flip
- This is known as **independence**
- We can consider the probability of a sequence of coin flips as the product of the PMF of each flip

$$\prod_{i=1}^n \theta^{k_i} (1 - \theta)^{1-k_i}$$

where $k_i = 1$ means coin i is heads

Putting this together

$$\mathcal{L}(\theta|X) = \Pr(X|\theta) \quad (4)$$

$$= \prod_{i=1}^N \theta^{I_H(X_i)} (1 - \theta)^{I_T(X_i)} \quad (5)$$

$$= \theta^{\#\{i: X_i=H\}} (1 - \theta)^{\#\{i: X_i=T\}} \quad (6)$$

- The parameter is the part of the model that determines the distribution
- If we had $Y \sim \mathcal{N}(\mu, \sigma^2)$ then the parameters would be μ, σ^2

Parameter

$$\mathcal{L}(\theta|X) = \Pr(X|\theta) \quad (7)$$

$$= \prod_{i=1}^N \theta^{I_H(X_i)} (1 - \theta)^{I_T(X_i)} \quad (8)$$

$$= \theta^{\#\{i: X_i=H\}} (1 - \theta)^{\#\{i: X_i=T\}} \quad (9)$$

- Here $\#A$ is the number of elements in the set A , and $I_a(X)$ is an indicator function (1 if $X = a$, and 0 otherwise)

Likelihood function

- Example: Suppose we flip the coin and get results H, H, H, T, T, H
- The likelihood is as follows:

$$\mathcal{L}(\theta|H, H, H, T, T, H) = \theta^4 (1 - \theta)^2$$

- Given the model and the data, the likelihood of $\theta = 0.5$ is: $\mathcal{L}(0.5|X) = 0.5^4 \cdot 0.5^2 = 0.5^6 = 0.015625$

Exercise

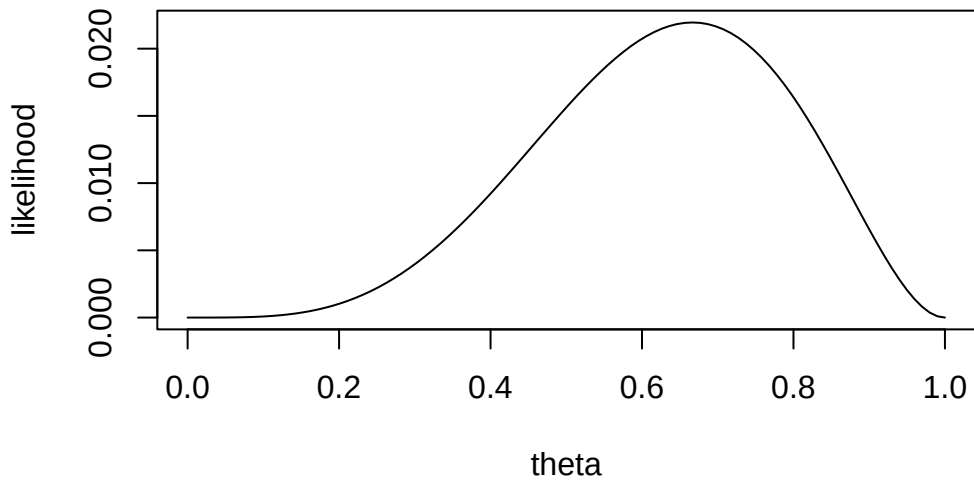
Write a function to compute the likelihood here, for a specific value of θ . The second argument to this function will be the number of heads, and the third will be the number of tails

Solution

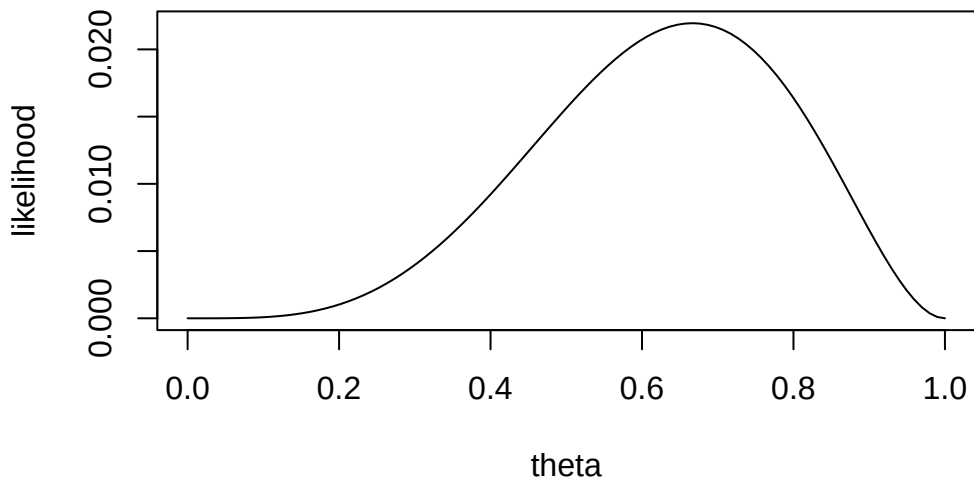
Likelihood function

Here's the likelihood as a function of θ :

```
xs <- seq(0, 1, length.out = 100)
ys <- (xs)^4 * (1 - xs)^2
plot(xs, ys, ylab = "likelihood", xlab = "theta", type = "l")
```



Likelihood



- The numbers on the y axis don't really matter, just the value of the parameter that gives the largest value

Exercise

- Modify the previous plot, instead plotting the log of the likelihood function on the y axis. What value of θ maximises the log likelihood? Explain this result

Solution

Maximum likelihood

- To do inference given a model, we find the parameter which maximises the likelihood
- This is essentially what we always do in every statistics problem, with some small modifications
- This is why finding the maximum of a function is so useful

We Saw this before

- Under linear regression, assuming Gaussian residuals, we have seen that the maximum likelihood solution is the ordinary least squares solution (the slope and intercept that minimize SSE)
- We can then extrapolate the maximum likelihood solution to the slope and intercept into regions of the x -axis for which we have no observations, and use the extrapolation to “infer” predictions

Maximum likelihood

- For the coin flip example, we have:

$$\mathcal{L}(\theta|X) = \theta^4(1 - \theta)^2 \quad (10)$$

$$\Rightarrow \frac{dL}{d\theta} = 0 \Leftrightarrow 0 = 4\theta^3(1 - \theta)^2 - 2(1 - \theta)\theta^4 \quad (11)$$

$$\Rightarrow \theta = 2/3 \quad (12)$$

- Minimizing likelihoods using derivatives is almost always easier working in log space

Exercise

- What is the maximum likelihood for the coin flip in general with n_T tails and n_H heads?
- Do this using the log of the likelihood, it's easier!
- Write a small R function to compute the maximum likelihood estimate given n_T tails and n_H heads
- We could also solve this problem using `optim` that we saw before!

Solution

- The log likelihood is

$$\log \mathcal{L} = n_H \log \theta + n_T \log(1 - \theta)$$

$$\frac{d\mathcal{L}}{d\theta} = \frac{n_H}{\theta} - \frac{n_T}{1 - \theta} = 0$$

Maximum likelihood

- Suppose we observe N coin flips. We're then asked if we want to play a game where we win \$1 if the coin comes up heads. What's the largest amount of money we should pay to play this game, if we want to expect to turn a profit? If we can answer this, what assumptions are we making? How is this inference?

Inference

- We've now learned a common inference technique: Define a probabilistic model, solve for the maximum likelihood parameter, use that parameter to do inference
- But, what if we cannot solve for the maximum likelihood by taking the derivative of the log, and setting it to zero? This will happen if the likelihood function is not convex (unimodal), when it is not continuous, or when there is no closed form solution

Example

- An example of a model with no maximum likelihood solution is a mixture model
- Remember that for linear regression, we defined the model by making assumptions about how the data were generated:
 - Suppose there is a true slope α , intercept β and variance σ^2
 - For each input point x_i , draw $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$
 - Then, “observe” the output $y_i = \alpha + \beta x_i + \varepsilon_i$

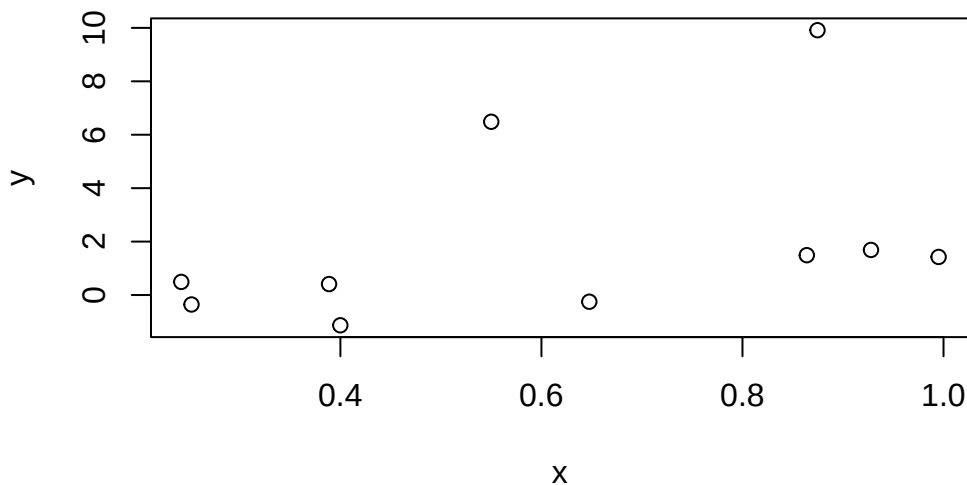
Example

So, here’s an example of “generating” linear regression data:

```
alpha <- 2
beta <- 1
sigma <- 3
xs <- runif(10)
ys <- alpha + beta * xs + rnorm(10, sd = sigma)
plot(xs, ys, ylab = "y", xlab = "x")
```

Example

So, here’s an example of “generating” linear regression data:



Mixture model

- Here's another example of a model (more advanced than linear regression and the “coin flip” model): A Gaussian mixture model
- Suppose there is a true number of components K , and for each component, a true mean μ_k and true standard deviation σ_k , and a true distribution over mixture components $\pi_1, \dots, \pi_K$ such that $\pi_k \geq 0, \sum_{k=1}^K \pi_k = 1$

More Details

- For each item i , draw $z_i \sim \text{Discrete}(\pi_1, \dots, \pi_K)$
 - Then, draw $x_i \sim \mathcal{N}(\mu_{z_k}, \sigma_{z_k}^2)$. “Observe” the data item x_i
- Here z_i is just an indicator of which of the K components x_i belongs to
- This determines the mean and variance of the distribution x_i is drawn from
- You only see x_i , not z_i or any of the μ, σ !

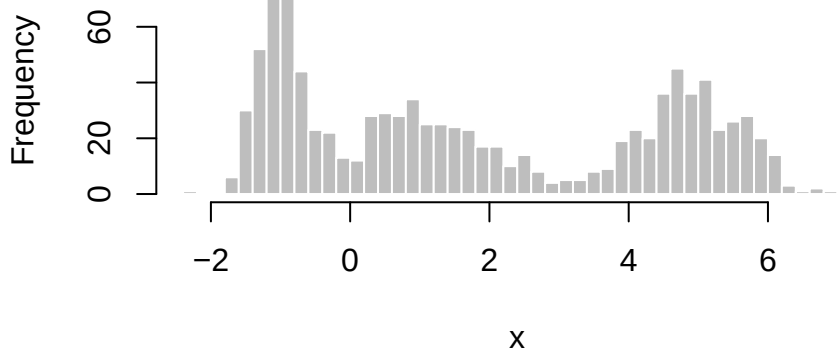
Mixture model

- Here's an example of “generating” data from a mixture model:

```
set.seed(240)
K <- 3
mus <- c(-1, 1, 5)      # means
sigmas <- c(0.1, 1, 0.5) # variances
varpi <- c(0.3, 0.35, 0.35)
N <- 1000
zs <- sample(K, size = N, replace = TRUE, prob = varpi)
xs <- rep(NA, N)
for (k in 1:K) {
  xs[zs == k] <- rnorm(sum(zs == k),
                      mean = mus[k],
                      sd = sqrt(sigmas[k]))
}
hist(xs, breaks = 50, col = "grey", border = "white",
     main = "1D Gaussian Mixture", xlab = "x")
```

Mixture model Data

1D Gaussian Mixture



Mixture model

- We can see a similarity to k-means: A dataset generated from a Gaussian mixture model looks like it would be ‘well modelled’ by k-means.
- But, there could be some mistakes at the edges of the clusters, due to some of the clusters being more compact than others

-
- But, we don’t have to use k-means. Just like in linear regression, where we find a maximum likelihood solution after assuming a model, we can “assume” a mixture model, and then find a solution for the parameters that yield high likelihood
 - But we can’t use calculus to solve for the derivative of the likelihood
 - We won’t show it here, but the likelihood function has many local optima, no “best” solution

Expectation maximization

- We can use a method called “expectation maximization” (EM). This is an iterative method (like k-means), that gets us mixture model parameters that are close to a maximum likelihood solution
- Details of EM are here: [MacKay. 2003. Section 33.7](#)

- You don't need to worry much about the theory behind it, but it works well for these types of problems

Intuition behind EM

- The procedure is kind of similar to k-means, where we repeatedly perform two steps
- E-step: Assign data to clusters (like in k-means)
- M-step: Update the parameters of the clusters (here they are normal means)
- Keep repeating this until convergence (you can prove never decrease a quantity)
- Much more general approach than k-means, very widely used in Statistics and Data Science

EM in R

- We run EM on the simulated mixture model data from the previous figure

```
library(EMcluster)

xmat <- matrix(xs, ncol = 1)

restarts <- 100
models <- vector("list", restarts)
lls <- rep(NA, restarts)

for (i in 1:restarts) {
  models[[i]] <- emcluster(xmat, simple.init(xmat, nclass = 3))
  lls[i] <- models[[i]]$llhdval
}

model <- models[[which.max(lls)]]
```

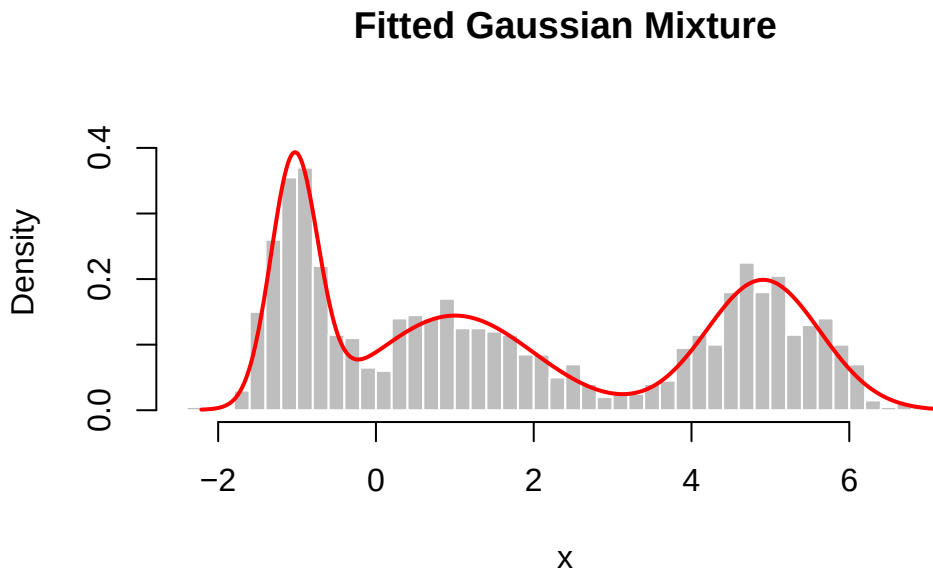
Fitting this model

- We said before there is no best solution for the mixture model
- This is just like k-means
- To find the best, we fit from 100 different random starts, choose the one with largest likelihood

```
model <- models[[which.max(lls)]]
```

- Each time finds the best model for that random start

The fitted model



Case study: London plague

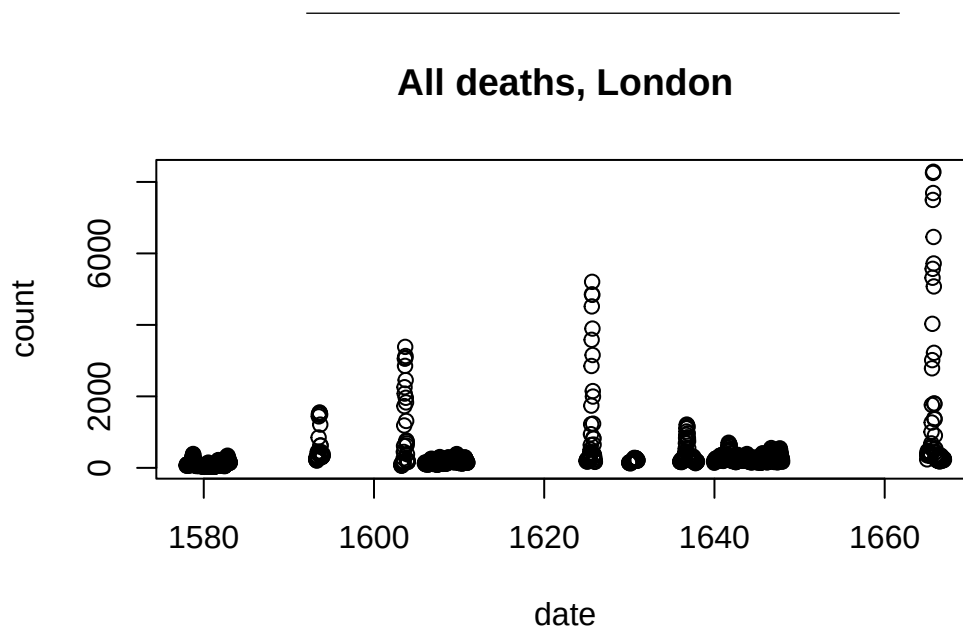
- The bubonic plagues struck Europe starting around the 14th century, and ending around the 17th or 18th century
- Regions were affected in waves lasting years
- We'll examine death counts from London, UK, and examine if we can get evidence for the idea of a “plague year” just from the death counts

Data source

- [Earn et al. Acceleration of plague outbreaks in the second pandemic. 2020.](#)
- This gives us death by all causes by day

```
library(lubridate)
load("londonbills.RData")
date <- ymd(londonbills$Date)
count <- londonbills$deaths_all_causes
date <- date[!is.na(count)]
count <- count[!is.na(count)]
plot(date, count, main = "All deaths, London")
```

- We want a histogram of yearly death



Data source

Records digitized by CANMOD (Canadian Network for Modelling Infectious Diseases)

The Diseases and Casualties this Week.



A Bortive	6
Aged	50
Ague	1
Apoplexie	2
Childbed	42
Chrisomes	11
Cold	1
Consumption	99
Convulsion	63
Cough	1
Droptic	22
Drownd at St. Martin in the Fields	1
Feaver	268
Fistula	2
Flox and Small-pox	4
Flux	1
Found dead in the Fields at St. Mary Islington	1

Frighted	1
Gowt	1
Grief	3
Griping in the Gurs	35
Jaundies	2
Imposthume	8
Infants	9
Kingsevil	2
Meagrome	2
Plague	5533
Purples	2
Rickets	10
Rising of the Lights	13
Rupture	1
Scurvy	5
Spotted Feaver	65
Stillborn	10
Stone	3
Stopping of the Stomach	6
Suddenly	1
Surfeit	36
Teeth	112
Thrush	3
Tifick	5
Vomiting	4
Winde	1
Wormes	12

Christned { Males — 68 }
 { Females — 78 }
 { In all — 146 }

Buried { Males — 3212 }
 { Females — 3248 }
 { In all — 6460 }

Plague — 5533

Decreased in the Burials this Week — 1837
 Parishes clear of the Plague — 7 Parishes Infected — 123

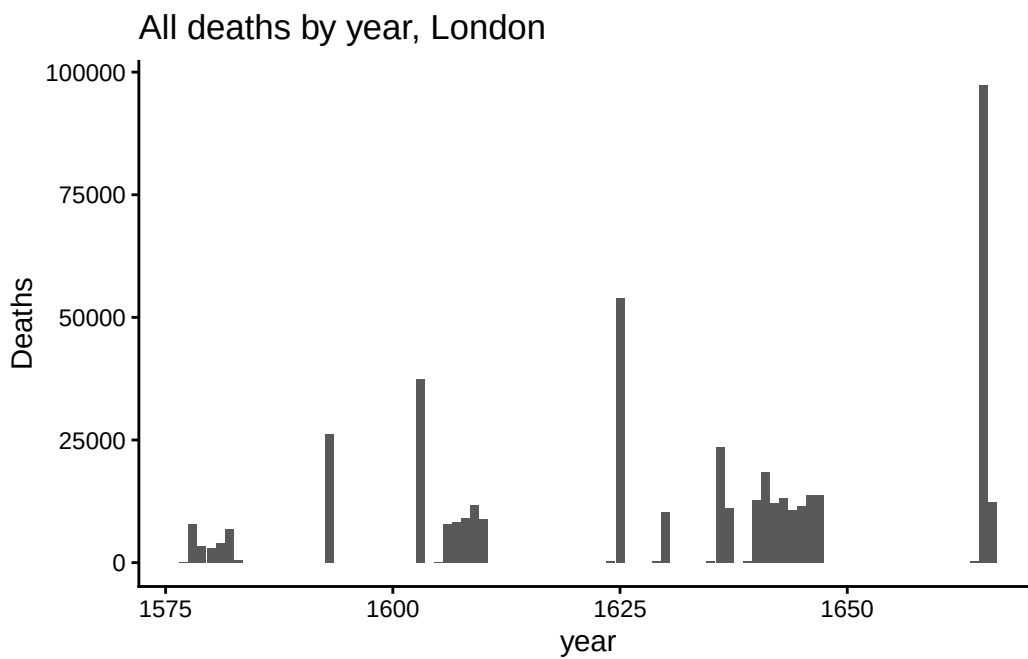
*The Assize of Bread set forth by Order of the Lord Mayor and Court of Aldermen,
 A penny Wheaten Loaf to contain Nine Ounces and a half, and three
 half-penny White Loaves the like weight.*

Bar Plot

- Bar plot, binned by year (note: this is not a histogram of yearly death)

```
data <- data.frame(date = date, count = count)
data$year <- lubridate::year(data$date)
binned <- data |>
  group_by(year) |>
  summarise(sum = sum(count))
ggplot(binned, aes(x = year, y = sum)) +
  ylab("Deaths") +
  geom_bar(stat = "identity")
```

Bar Plot



- Discussion: What about boundaries with the missing data?

Bar Plot

- Binned by decade: Also not a histogram

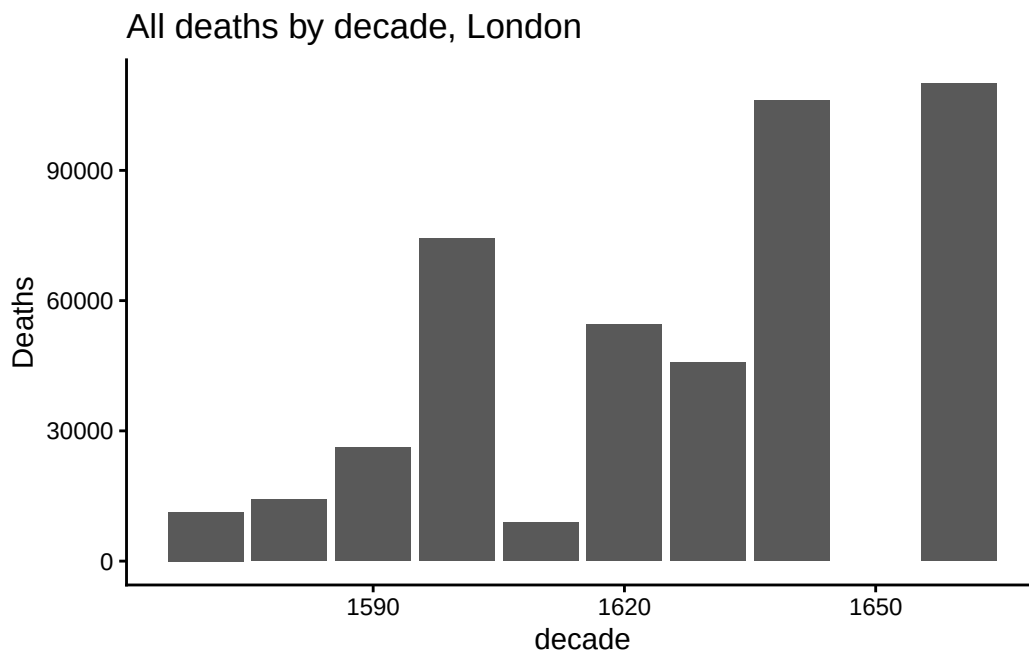

```

floor_decade <- function(value) {
  return(value - value %% 10)
}
data$decade <- floor_decade(data$year)
binned <- data |>
  group_by(decade) |>
  summarise(count = sum(count))

ggplot(binned, aes(x = decade, y = count)) +
  ylab("Deaths") +
  geom_bar(stat = "identity") +
  ggtitle("All deaths by decade, London")

```

Bar Plot



Histogram

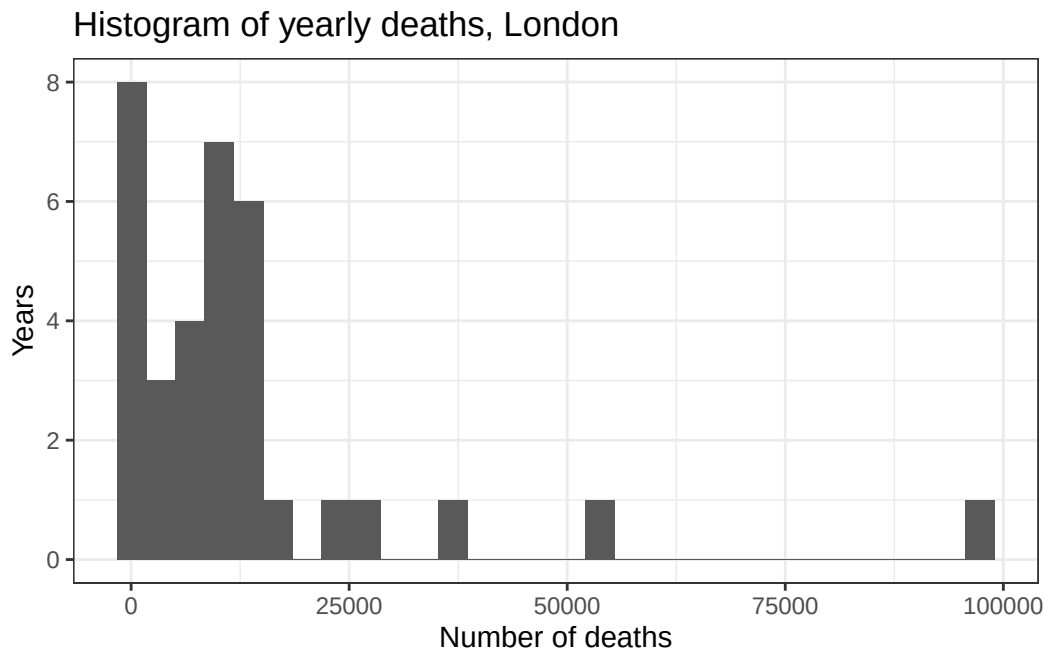
- Think of a histogram as almost like a one dimensional scatterplot

```

binned <- data |>
  group_by(year) |>
  summarise(count = sum(count))
ggplot(binned, aes(x = count)) +
  ylab("Yearly deaths") +
  geom_histogram()

```

Histogram



GMM on yearly deaths

- How many clusters are there in yearly death? If there are more than one, we can see a “difference” between different “types” of years, in terms of death rate. For $K = 2$:

```

restarts <- 100
xs <- data.frame(count = data$count)
m2s <- vector("list", restarts)
l12s <- rep(NA, restarts)
for (i in 1:restarts) {
  m2s[[i]] <- emcluster(xs, simple.init(xs, nclass = 2))
  l12s[i] <- m2s[[i]]$llhdval
}

```

```

}
m2 <- m2s[[which.max(ll2s)]]
print(m2$Mu)

```

```

      [,1]
[1,] 200.1383
[2,] 1831.2050

```

GMM on yearly deaths

- For $K = 1$:

```

m1s <- vector("list", restarts)
ll1s <- rep(NA, restarts)
for (i in 1:restarts) {
  m1s[[i]] <- emcluster(xs, simple.init(xs, nclass = 1))
  ll1s[i] <- m1s[[i]]$llhdval
}
m1 <- m1s[[which.max(ll1s)]]
print(m1$Mu)

```

```

      [,1]
[1,] 339.1172

```

Results

K	Log.likelihood
1	-10616.139
2	-8584.244

- Model 2 ($K = 2$) has a larger large likelihood. But, there's a problem: We usually can't directly compare "point estimates" of the likelihoods of models that have different numbers of parameters (Occam's razor)

Results

- One way to take into account Occam's razor for model comparison is through use of "information criteria" such as "Bayesian information criteria"
- $BIC = M \log N - 2 \log \mathcal{L}$
 - Here N is the number of data points
 - M is the number of parameters in the model (in this case for $K = 1$ this is 2, for $K = 2$ this is 5)
 - A lower BIC means the model is a better fit

Computing this

```
N <- nrow(xs)
results <- data.frame(
  K = c("1", "2"),
  BIC = c(
    2 * log(N) - 2 * m1$llhdval,
    5 * log(N) - 2 * m2$llhdval
  )
)
kable(results, row.names = NA)
```

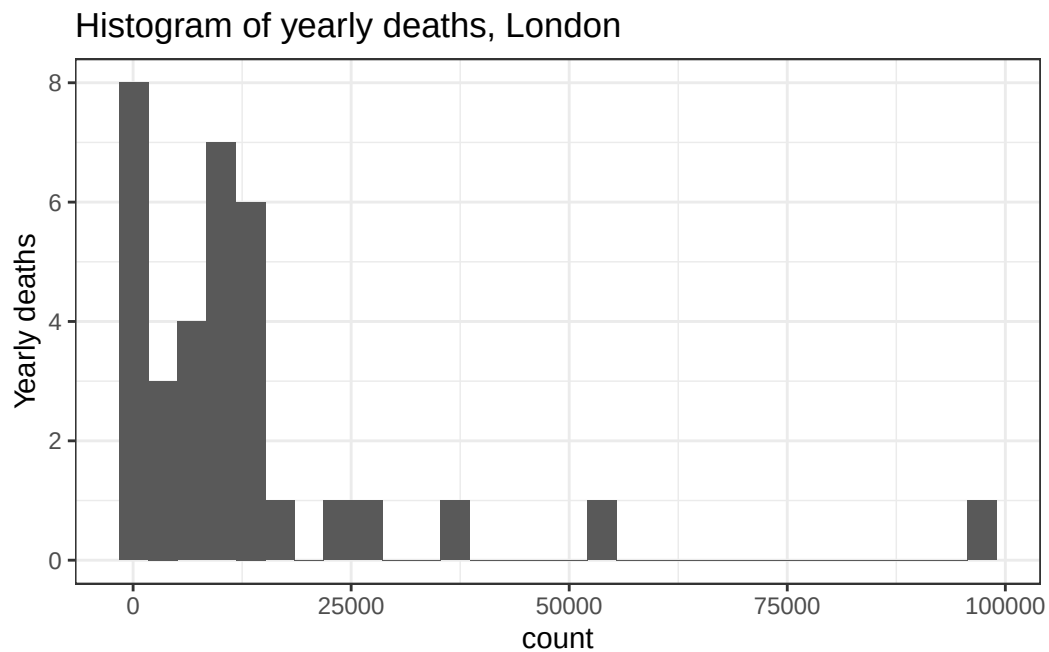
K	BIC
1	21246.67
2	17204.46

Conclusion

- We model the yearly death in London, UK from 1577 to 1666 using M1: A mixture model of two Gaussian distributions. And M2: A single Gaussian distribution. Model M2 is a better fit (means 200.1, 1831.2), indicating that it is unlikely that yearly deaths in London, UK for this period are Gaussian with a single fixed mean
- This makes sense when we looked at the histogram
- There are many other details about comparing these models also

The Histogram again

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Some Questions

- To what extent can we conclude from this that there are plague years in these data?
- We're not modelling the "severity" of the plague year (it either is or it isn't)
- We're making Gaussian assumptions when Poisson would be better
- We're not using covariates such as population size. And we're introducing errors at the "boundaries" of the missing data